

STUDIES ON PRODUCTION OF GREEN HYDROGEN

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Introduction

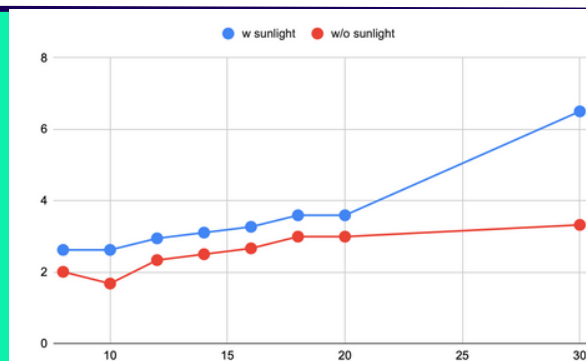
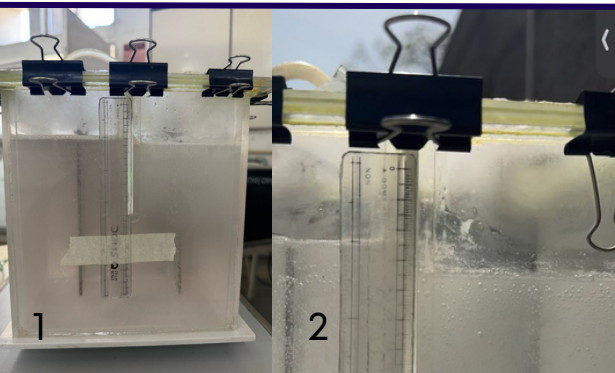
- Project explores the sustainable production of green hydrogen via a hybrid method combining photocatalysis and electrolysis

Research Objectives

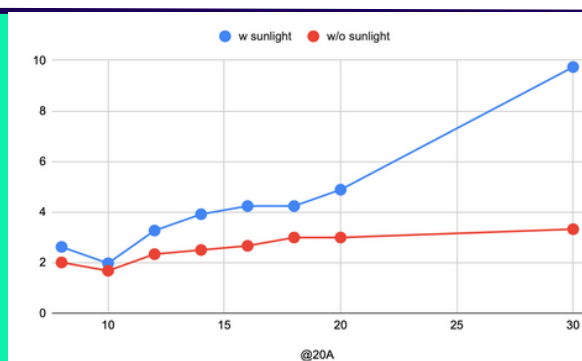
- Investigate green hydrogen production using TiO_2 photocatalysis.
- Optimize TiO_2 -coated electrodes for efficiency.
- Reduce energy consumption in electrolysis using solar energy.
- Develop a cost-effective method for sustainable hydrogen generation.

Methodology

- Volume of water displaced by Hydrogen is used to calculate the moles of gas produced
- Variables such as voltage, current, and sunlight exposure are used for comparison



% Increase for TiO_2
36%



% Increase for
 ZnO-TiO_2
62%

Results

- Hydrogen evolution rate increased under solar irradiation.
- TiO_2 photocatalysis reduced the required external energy.
- Higher efficiency observed at increased sunlight intensity.
- Results indicate the feasibility of hybrid solar-electrolysis approach.

Conclusion

- This study demonstrates the viability of integrating photocatalysis with electrolysis for green hydrogen production.
- The optimized TiO_2 electrodes enhance hydrogen yield while reducing electrical energy input, making the process more sustainable and scalable.

References

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- Y. Yao, X. Gao, and X. Meng, "Recent advances on electrocatalytic and photocatalytic seawater splitting for hydrogen evolution," *Int J Hydrogen Energy*, vol. 46, no. 13, pp. 9087–9100, Feb. 2021, doi: 10.1016/j.ijhydene.2020.12.212.
- S. Shiva Kumar and V. Himabindu, "Hydrogen production by PEM water electrolysis – A review," 2019, doi: 10.1016/j.mset.2019.03.002.